Brain tumour segmentation *exercises*

Agentic Medical AI Lab

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Four rules

They apply to every exercise on the following slides.

1. **Explore first.**

   Then work in *plan mode*: read the plan, edit it, and only then execute.

2. **Make figures generously.**

   Every number needs one, and figures are cheap. Never accept a number without one.

3. **Decide what good looks like**

   before you produce a result — not after you see one.

4. **Report what you found**

   and how you got it — not just the prompt, and not what you hoped for.

The data

Use the standard 60-case dataset as your default. If processing time starts to hurt, switch to the tiny 15-case set — say so when you report a number, because the number depends on it.

60 cases `tier_standard.txt`
DEFAULT

15 cases `tier_tiny.txt`
FALLBACK IF SLOW

Each case

Four channels, one mask

T1	native
T1ce	contrast-enhanced
T2	native
FLAIR	fluid-attenuated
mask	what you are predicting

Keep a diary

Start noting your own personal experience

- Regularly write a few lines. Not what you did — what you *noticed*.
- Eg. where an agent was confidently wrong, and what in your prompt let it be.
- The correction you had to make twice. That one is a rule, not an accident.
- What you had to insist on: tests before claims, a figure with every number, split by case.
- What worked so well you want it by default from now on.

By Tuesday evening you should have 10–15 lines. Fold them into your own `CLAUDE.md` or `rules.md` — that file, not the code, is what you take home.

rules.md

```
## my rules — day 1

- state the metric AND the baseline
  before running anything.

- it defaults to slice-level splits.
  say "split by case" every time.

- it reports the mean. ask for the
  failures — the mean hid case_026.

- no number without a figure.

- "done" from the model is a claim,
  not a result. make it show me.

- plan mode: it starts editing
  before i have agreed to anything.
```

E1 Set up

- Install Claude Desktop and log in — you need a Claude Pro, Max or Team account. Alternatively use OpenAI Codex. For a free option use OpenCode free tier.
- Have a GitHub account ready. You don't need it to run locally, but you do if you switch to the Codespace fallback.
- Check you have Python 3.11 or 3.12. Install it if not.
- Clone this repository, then run `python verify.py` until it says ALL GOOD.
- Stuck for 15 minutes? Try the Codespace to run in cloud instead of local.

E2 Inspect the data

- Ask Claude to describe the dataset from the files alone: how many cases, which channels, what exactly is being predicted, how big is a voxel. Then verify one of its claims yourself.
- Load one case. Print the shapes, and what fraction of the brain is tumour. Is that fraction plausible?
- Now do it for every case at once: tumour volume, tumour percentage, and how much they vary between cases. Put results in a summary table.

E3

Build a viewer

- Write your specification on paper first.
- Make a plan with ‘plan mode’. If the plan does not match your specification, edit it before you let it run.
- Build
- Choose a case, move through slices, see all four channels at once, turn the tumour mask on and off.
- Check the mask lines up with something real. A viewer showing the wrong slice runs perfectly.
- Look at `case_039`, then at `case_026`.

E4

Is there signal?

- Predict first, in writing: which channel separates tumour from brain best? Commit to an order.
- Plot tumour against non-tumour intensity for all four channels. Ask for a number and a picture.
- Plot every channel against every other, tumour voxels coloured. Do `case_039`. Then do `case_026`.
- Put the two figures side by side. Were you right?
- Now plot the mean brain intensity of ten different cases. What does that imply for any rule written in raw units?

E5 One number

- On `case_039`, find the FLAIR cutoff that best separates tumour from brain. Define “best” before you search.
- Show the score against the cutoff, and the ROC curve. Mark the operating point you chose.
- Map the results in three colours: correct, false alarm, missed.
- Predict the median Dice. Then apply that same cutoff, unchanged, to every case in the dataset.
- Now express the cutoff in standard deviations above each brain's own mean, and repeat.
- Plot both. What happened, and why?

! *Your accuracy may be above 95%. What would “predict nothing anywhere” score?*

E6 Use all four channels

- Fit a voxel-level logistic regression on the four normalised channels. Tumour is ~1% of voxels — balance the training sample.
- Read the four coefficients out loud. Do they match what you would expect given your knowledge about the four brain map types? If not, who is wrong.
- Your best probability cutoff will not necessarily be 0.5. Find it, and explain the discrepancy.
- Compare against E5 case by case. Is the gain large, or merely positive?
- How sure are you of that difference? Would 15 cases have been enough to claim it?
- Plot Dice against tumour volume. What explains the failures?

E7 Learn the features

- Predict first: what kind of mistake should a model that can see the neighbourhood fix?
- Train a small 2D U-Net. Split by case, not by slice since slices from one patient are near-duplicates.
- Compare with E6 on held-out cases and look at the error maps. What changed?
- Now leave the architecture alone and add augmentation and test-time augmentation. Re-measure.
- How much did you gain *without touching the model*?
- Note the image size you trained at, then re-check at full resolution. Does the number survive?

! *Before you leave: write down the single number you would put in an abstract, and the sentence around it. Keep it.*